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Thermostat Controller Hardware Evaluation:
Comparative Analysis and Production Recommendation

SysTec's prototype thermostat was developed on the Raspberry Pi 4B, which provided a flexible development environment well-suited for rapid prototyping. However, transitioning to a production product requires evaluating hardware architectures against three core business requirements: peripheral support, cloud connectivity via Wi-Fi, and sufficient Flash and RAM to support the application code. Three architectures were evaluated: the Raspberry Pi 4B, the Microchip WFI32E04, and the NXP Kinetis K64F.

Table: Hardware Architecture Comparison

Requirement	<i>Raspberry Pi 4B</i>	<i>Microchip WFI32E04</i>	<i>NXP Kinetis K64F</i>
<i>Peripheral Support: I2C, GPIO, UART, PWM</i>	Yes -- 6x UART, 6x I2C, 28x GPIO, up to 2x PWM channels	Yes -- 2x I2C, 3x UART, up to 62 GPIO, PWM via Output Compare modules	Yes -- 3x I2C, 6x UART, up to 100 GPIO, hardware PWM via four FlexTimer modules
<i>Wi-Fi / Cloud Connectivity</i>	Yes -- integrated 802.11 b/g/n/ac	Yes -- integrated 802.11 b/g/n	No, requires external module
<i>Flash</i>	External Flash via microSD	2 MB	1 MB
<i>RAM</i>	4 GB LPDDR4 RAM	512 KB SRAM	256 KB SRAM

All three architectures support the peripherals used in the prototype, including I2C, UART, and GPIO. However, the architectures differ significantly in how they meet the remaining requirements. The K64F is eliminated from consideration due to its lack of integrated Wi-Fi, which directly conflicts with the requirement for cloud connectivity. An external Wi-Fi

module could theoretically address this gap, but doing so introduces hardware complexity that makes integrated alternatives preferable. The Raspberry Pi 4B includes integrated Wi-Fi and fully supports all required peripherals but presents a different problem: the specs significantly exceed the computational demands of the application. Its quad-core CPU and 4GB of LPDDR4 RAM are compute resources that a thermostat application won't utilize, and its per-unit cost at production scale makes it cost-prohibitive relative to scaled-down alternatives.

The Microchip WFI32E04 is the recommended architecture for production. It satisfies all three business requirements: it supports I2C, UART, GPIO, and hardware PWM via dedicated Output Compare modules. It includes integrated Wi-Fi with WPA2/WPA3 security, and its 2 MB of Flash and 512 KB of SRAM are sufficient to support the thermostat application. The WFI32E04 also offers hardware-based security well-suited for a cloud-connected product, including a crypto engine supporting AES, 3DES, SHA, and HMAC with a True Random Number Generator, and TLS v1.3 support with hardware acceleration for encrypted server communication. As a purpose-built IoT microcontroller, it carries none of the overhead of a Linux-based system, operates at significantly lower power, and is cost-effective.

References

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